Session-I: Introduction to Farmer Water Schools (FWS) – June 3rd week

Program schedule

S. No	Time	Content	Method	Materials	Expected outcomes
1	10 min.	Introduction of participants and sharing Expectations.	Dyad	Picture charts of different fruits/ vegetables, and box.	Participants are introduced to one another.
2	30 min.	FWS Objectives.	Large group discussion	Charts, markers and pictures.	Participants list their expectations from year-long FWS and compare with FWS objectives.
3	60 min.	Pre-Ballot Box Exercise.	Demonstration	Charts, markers whistle, card boards, nails, long sticks, knife, thread, and ballot boxes.	Participants participate in BBE.
4	25 min.	Elements of farmer decision making process.	Large group discussion	Charts, markers, and visuals.	Farmers list the steps involved in decision making process.
5	20 min.	Norm setting.	Story and large group discussion	Pictures, charts, markers, and script for the story	Participants set norms for the year long FWS sessions.
6	25 min.	Group formation.	Game and large group discussion	10 stones of medium size, different varieties of stones, charts, markers and board.	• Participants identify the Importance of group formation • Participants form as small groups to learn and share their experiences in FWS sessions.
7	25 min.	FWS Pledge.	Pledge	Charts and markers	participants discuss and decide the content for FWS Pledge
8	10 min.	Group dynamics-Importance of peoples' institutions.	Game	Match sticks.	Participants understand the importance of institutions.
9	15 min.	Introduction to short studies.	Large group discussion	Charts and markers.	Participants are introduced to the concept of short studies.
10	20 min	Short Study-Green manure.	Large group discussion	Handouts.	Participants discuss the advantages of green manure.
11	10 min.	Evaluation.	Large group discussion.	Charts and Markers	Participants share their opinions on the conduct of FWS sessions and methods used in the session.

Learning objectives

By the end of the session, participants will be able to:

- Introduce one another,
- Discuss objectives of Farmer Water School (FWS),
- Participate in Pre-Ballot Box Exercise,
- Discuss elements of farmer decision making,
- Form small groups and set norms for smooth and effective conduct of FWS sessions, and
- Discuss the importance of people's institutions.

Process

1. Introduction of participants and sharing of expectations Time: 10 min.

Objective : Participants introduce one another.

Content : Participant introduction.

Method : Dyad technique

Materials : Pictures of fruits, and box.

Process

- Collect two pictures of each variety of fruit. The total number of pictures should match the number of participants (Ex: For 30 participants collect pictures of 15 pairs of fruit).
- Put these pictures in a box and ask each participant to pick one picture from the box.
- Encourage participants to lookout for the other participant having the same picture and ask them to form pairs.
- Ask the participants in each pair to introduce themselves to their partner. Participants should use the following information to guide the introductions:
  - Name,
  - Village and hydrological unit, and
  - Position in GMC/HUN (spell out)or any other institution.
- Later, ask each participant in the pairs to introduce his/her partner to the large group.
- Continue the introductions till all the participants introduce one another.



Pictures of fruits and vegetables in pairs

2. FWS objectives Time: 30 min

Objective : Participants discuss FWS objectives.

Content : FWS objectives.

Method : Large group discussion

Materials : Charts, markers and pictures depicting Indian groundwater scenario.

Process

- Ask farmer participants to reflect for a minute on issues around water availability and agriculture that they wish to discuss/learn during the year-long Farmer Water Schools.
- Use the following visuals to initiate discussion on the present groundwater scenario.




Results of water levels depletion

- Keeping in view the existing groundwater situation, encourage the participants to share the issues that need to be discussed during the year long FWS sessions that can provide us a good understanding on groundwater science and water management techniques.
- Establish a norm that all ideas are welcome (no idea is a wrong one) and others should not judge an idea as it is being shared.
- Note the list of participants' ideas on a chart.
- Ensure that each participant gets a chance to share their thoughts.
- After all ideas have been shared, open the floor for discussion.
- In the large group, review the list of ideas and sequence the events.
- Thank the participants for the inputs and share the FWS objectives.
- Request the participants to compare their expectations with the FWS objectives and ask them to suggest required additions if any.
- Following these discussions, finalize the objectives for the year long FWS sessions.

The FWS objectives are as follows:

- Empower farmers with knowledge and skills in measuring recharge & draft,
- Sensitize farmers on the need for collective action,
- Sharpen the farmers' ability to make critical and informed decisions on crop plans,
- Sensitize farmers on new ways of thinking and resolving issues.

3. Pre Ballot Box Exercise (BBE) Time: 60 min.

Objective : Participants participate in Ballot Box Exercise.

Content : Ballot Box Exercise.

Method : Demonstration

Materials : Charts, markers, card boards, nails, long sticks, knife, thread, ballot boxes, and whistle.

Process

- Share the purpose of Ballot Box Exercise (BBE).The purpose of BBE is to know participants' awareness on groundwater, types of soils, rocks of HU and other water use efficiency methods. This will help determine topics that need focused attention during FWS sessions.
- After sharing the purpose of BBE, assign a registration number to each participant.

- Give each participant 30 ballot slips (number of slips should be equal to the number of questions) bearing his/her registration number.
- Ask each participant to lineup in front of one cardboard station (i.e. one participant in front of one cardboard station).
- Each cardboard station has a question and three options (yes, no, no idea).
- Beneath the question, each option has a cover.
- Ask the participants to read the question, choose the appropriate option and drop the ballot into that cover.
- On blowing the whistle, participants move to the next station to read the question and cast their votes for the question.
- Participants exercise their ballot at each station.



Tips to facilitator

- Help participants distinguish between a cardboard station, a ballot, and a ballot box.
- Before the conduct of the exercise, explain all the instructions to the participants.
- Organizing a mock demonstration before the actual conduct of the exercise will help participants.
- Assist illiterate or semi-literate farmers by reading the question and options.
- Only one participant should stand near one cardboard station.
- Do not allow two or more persons to crowd near one cardboard station.

Analysis of BBE Results

- First take out chits from 'No Idea' pocket, count them and record on a chart.
- After this, go to the pocket with least number of chits, count them and record on a chart.
- Finally record the third one by deducting the sum of the two from the total participants (There is no need to count the chits in the third pocket).
- Categorize the questions based on similarity.
- Calculate cumulative right/ wrong/ no idea question-wise and individual farmer-wise.
- Calculate percentages.
- Based on the results, determine topics that need focused attention during FWS.

4. Elements in farmer decision making process Time: 25 min.

Objective : Participants discuss steps in decision making process.

Content :

- Steps in farmer decision making process.
- Farmer's problems around groundwater and agriculture.
- Four core components of Farmer Water School.

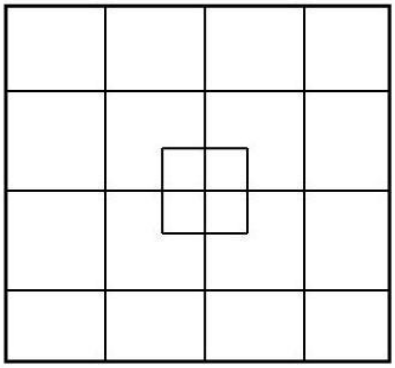
Method : Large group discussion

Materials : Charts, markers and visual having squares.

Process

Elements of farmer decision making process

- In a large group, show the following visual:



- Ask participants to observe the picture for 2 to 3 minutes and then count the number of squares in the picture. Do not lay any norms for participation in this activity.
- Later, ask a few participants to share their finding.

Tips to facilitator

- Few participants count quickly and find fewer squares.
- Others take longer time, help each other and find more.
- The more they analyze and discuss, the more clearly they arrive at the right answer.
- The correct answer is 35 squares.

- Ask them to describe how they arrived at their conclusions. List the steps involved in arriving at the said conclusions. The steps include:
  - See – observe.
  - Discuss with others.
  - Brainstorm – Think.
  - Make a decision.

Decision making process and groundwater management

- To enable better understanding, ask participants to list problems related to groundwater and agriculture.
- Take any one of the problems and ask the following question: "Assume that you do not have enough water in your borewell to irrigate the crop. How would you approach this problem and find a solution?"
- Record the participant's responses on a chart paper.
- Summarize the discussion by keeping the participants responses in mind and share that, you will 'discuss' with your fellow farmers and seek their advice. You might get various suggestions from them like shift of crop, going for a new borewell, purchasing water from

other sources etc. After gathering information, you 'think' which is the best possible way to overcome the problem, and finally you take the appropriate 'decision' and implement the same.

- List the steps involved in the decision making process. They are:
  - See –observation.
  - Discussion- discussion with others.
  - Think – analysis.
  - Decision and implementation - decision making.
- Ask the participants if we could use these steps to develop strategies to address problems around groundwater availability and usage?
- Encourage participants to respond and reach consensus.

5. Norm setting Time: 20 min.

Objective : Participants set norms for smooth conduct of FWS sessions.
Content : Setting norms.
Method : Story telling and large group discussion.
Materials : Pictures, charts, markers and script of father and two son's story.
Process

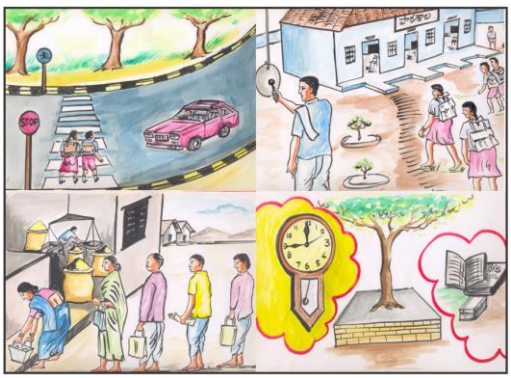
- For effective and smooth conduct of Farmer Water School sessions, and to help the facilitators in the conduct of FWS sessions, encourage the participants to lay down the norms.
- To facilitate this, use the following pictures and generate a discussion on the norms observed in the pictures. Like wise, list the norms for effective conduct of Farmer Water Schools that evolve from the discussion on a chart paper.

Norms at various places

- Another option to set norms is to narrate the following story

Story: Father and two son's

There lived a man with his two sons. Due to financial problems, he could send only his elder son to school and kept the younger son with him to help in farm activities. On one occasion, he had to send his elder son to a nearby town to meet an official. So, he asked the younger son to go to the school in place of his elder brother. But the class teacher rejected this boy and sent him back.



After sharing the story, ask the following questions:

- What have you learnt from the story?
- Why was the younger son not allowed to sit in the class?

- List participant responses and lay down the norms for the effective conduct of Farmer Water Schools.
- Write the norms on a chart and read them aloud.

Tips to facilitator:

Encourage the participants to set the norms regarding:

- Time management.
- Transportation.
- Venue, responsibilities of host teams in the conduct of session.
- No substitute should attend the FWS session.
- Regular attendance.
- Discuss the learning of FWS session in GMCs.

6. Group formation Time: 25 min**.**

Objective : Participants form into small groups.

Content :

- Why small groups?
- Formation of small groups.

Method : Game and large group discussion

Materials : Medium size stones (ten in number), different varieties of stones.

Process

- Make the participants to play the game with 10 stones.

Game: Stone Game

- Take 10 stones of medium size.
- Divide the participants into two groups with five participants only in one group and the rest (say 25 participants) in the second group.
- Keep the 10 stones (each weighing two kilos) at one place.
- Ask the first group (five members) to move away a little and arrange the stones in order.
- Next ask the second group (25 members) to do the same a little away from the first group using the same stones.

- Ask the following questions after the game:
  - What have you observed in the game?
  - What is the difference between first group and second group?
  - Write participants responses on a chart paper and share the following advantages of having small groups.

A. Every member gets an equal opportunity to share their experiences and can articulate easily with other members in the group.
B. Every participant can improve his/ her presentation skills.
C. Enables mutual learning and promotes team work.
D. Increased camaraderie.

Encourage the participants to play the following game and form small groups.

Game: Different kinds of stones

- Take different varieties of stones available in the specific HU.

- Ensure that the number of stones is equal to the number of participants.
- Ask each participant to pick one stone.
- After the exercise, any two participants getting the same type of stone are grouped together.
- Encourage the participants in each group to read out their names before the audience.
- Encourage each group to name their group and identify a leader.

All the group leaders identified are responsible for the following:

- To ensure that all the members in that specific group are present for the session.
- To lead the discussion and ensure that there is an equal participation from every group member.
- To encourage every group member to present the groups work in large group.
- Before concluding the discussion, share that the groups will remain as a cohesive learning unit for the entire duration of Farmer Water Schools.

7. FWS Pledge Time: 25 min.

Objective : All the participants discuss and decide the content for FWS Pledge

Content : FWS pledge Content

Method : large group discussion

Process

- Ask the participants why school children take a pledge while beginning their regular day.
- Elicit responses from the large group and state that a pledge is to inculcate a discipline in our mind and enables us to take a work with more seriousness.
- State that it will be like promising ourselves that we will save our groundwater resources together.
- End the discussion by posing a question to the large group, if it will be good to start the future FWS sessions with FWS pledge.
- Encourage the participants to discuss what would be the ideal content in FWS pledge.
- Finalize FWS pledge.

8. Group dynamics- Importance of peoples' institutions Time: 10 min.

Encourage the participants to play this game. The purpose of this game is to discuss the importance of a group who has common interest to solve their issues.

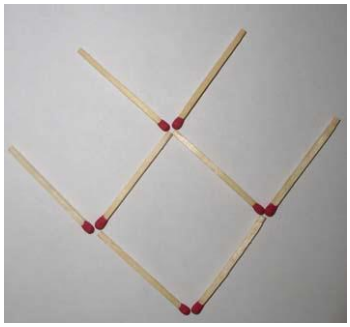
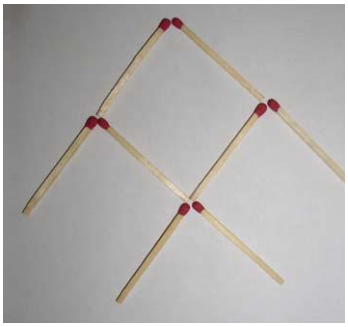
Game: Fish Game

Use eight match sticks to symbolize a down ward looking fish as shown in the picture given below. Ask participants to move just three matchsticks to change the direction of the fish so that it is facing upwards.

- After the game, ask participants the following questions:
    - What is your inference regarding the downward and upward direction of fish?
    - How did you change the direction of fish?
    - Can we relate these three match sticks to collective action, Unity and Mutual assistance which are important for a people’s institution?

Message: The three sticks represent the three things which one needs to progress in life:

- Collective action,
- Unity, and
- Mutual assistance.

9. Introduction to Short Studies Time: 15 min.

Objective : Participants discuss the problems they have been facing.

Content : Introduction to Short Studies.

Method : Large group discussion

Material : Charts and markers.

Process :

- In a large group, ask participants the following question.
    - You have been growing crops for many years. What are the major problems you people are facing?
- List the participant's responses on chart paper. Problems could be pest related, water related or related to marketing.
- This list should be used to provide assistance or design short studies at various stages of FWS.

10. Short study: Green manures Time: 20 min.

Objective : Participants discuss:

- Types of natural manures
- List of green manures
- Feasibility of application of green manures in the field

Content : Green manure crops

Method : Large group discussion

Material : Charts and markers

Process

- In a large group, ask participants the following question: "What are the ways of increasing soil fertility?"

- Ask the following question in large group: "What are the characteristics that should be present in a plant to be used as a Green manure crop?" Record the participant responses on chart paper.

Tips to the facilitator

Characteristics of green manure crops

- Have profuse leaves and rapid growth early in its life cycle.
- Have abundance and succulent tops.
- Be capable of making a good stand on poor and exhausted soils.
- Have a deep root system.
- Legume with good nodular growth habit.
- Examples of Green manure crops are oats, rye, mustard, clover, fenugreek, fava beans, sun hemp, alfalfa, azolla etc.

- Ask participants to list the types of plants that can be used as 'green manure'.
- Assign each group a specific plant and ask them to discuss in their groups:
- Discuss the process of converting the plants into green manures?
- What are the benefits of using 'green manures'?
- Later, ask each group to report out their discussions to the large group.

Tips to facilitator

Benefits of green manure

- Green manure offers an inexpensive way of improving crop yields and takes little effort.
- Recycle and add organic matter to soil.
- Leguminous plants help in supplying nitrogen to soil.
- Improves soil structure by letting more air and also improves drainage.
- Prevent soil erosion.
- Help in controlling weeds.
- Increases absorptive capacity of soil.
- Increases biochemical activity in soil by becoming food to micro-organisms.
- Helps in increasing crop yield.
- Can be used for mulching

11. Evaluation Time: 10min.

- Ask the following questions to evaluate the day's learning. Pose the questions to the large group:
    - What did you like about today?
    - Do you have any suggestions on the topics discussed today?
    - Do you think the steps in decision making process are relevant in daily life? How?
    - How will participation in FWS improve collective efforts for sustainability?
- Note the participants feed back and thank everyone for their active participation.